Week 06 teacher guide

Expressions and equations

Keep this guide beside PowerPoint slides 316-376. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

Day 036. Saturday | pages 141-145 | slides 316-333
Expressions and equivalent forms

Day 037. Sunday | pages 146-155 | slides 334-352
Equations and related quantities

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Weekly check

This week’s idea

Expressions name quantities. Equivalent forms describe the same quantity. Equations can be solved by keeping both sides equal.

Use the built-in Booster review where it appears in the day’s work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook’s Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Expressions and equivalent forms

Day 036, pages 141-145 | PowerPoint slides 316-333

Goal: Write expressions and show why two forms have the same value.

Set out: square tiles, grid paper, pencil.

Check first: Multiplication, fraction products, and grouping symbols.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Draws repeated groups before writing 5n. Why it matters: Connects a variable to a quantity. Tutor move: Ask what changes when one sheet costs more.

Partitions the rectangle and counts both areas. Why it matters: Explains distribution visually. Tutor move: Ask which area matches each term.

Uses either repeated addition or a rectangle correctly. Why it matters: More than one model can show equivalence. Tutor move: Accept both and compare the matching groups.

Likely misconceptions

Writes $4^2 = 8$. Why: Reads the exponent as multiplication by two. Tutor move: Write two factors of 4, then compare.

Writes $4(x + 3) = 4x + 3$. Why: Loses the four groups of 3. Tutor move: Shade the numeric part in all four rows.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 316. Week 6, Saturday

Topic: Expressions and equivalent forms
Student: Watch and discuss

Say: Write expressions and show why two forms have the same value.

Ask: What do you already know about this idea?

Expect: Multiplication, fraction products, and grouping symbols.

If stuck: Ask what changes when one sheet costs more.

Slide 317. Today

Topic: Expressions and equivalent forms
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.

If stuck: Read the learning goal together.

Slide 318. Recall | 4 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 141 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 0.65 dollars per marker; 2: 6; 3: 12; 4: 60 shells

If stuck: Give time to draw before giving a hint.

Slide 319. Recall: answers | 3 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 141 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 0.65 dollars per marker; 6; 12; 60 shells

If stuck: Rebuild one missed item with its model.

Slide 320. Expression from a story | 8 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 142 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: What quantity does the letter represent? How many equal groups are there?

Expect: P1: 5n; $10; $20

If stuck: P1: Draw one group and label its unknown amount. Repeat the whole group before writing the expression.

Slide 321. Evaluate using order of operations | 4 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 142 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: P2: What number replaces the letter? Which operation happens first? P3: How many equal factors does the exponent name?

Expect: P2: 17; P3: 16; 8; 27; $\frac{4}{9}$. 4^2 and 4×2 have different meanings.

If stuck: P2: Rewrite the expression with the given value. Keep multiplication together before adding. P3: Write the repeated factors before multiplying. Compare that with multiplying the base by the exponent.

Slide 322. The model | 4 min

Topic: Expressions and equivalent forms
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Four rows each contain x + 3.

If stuck: Ask what changes when one sheet costs more.

Slide 323. Distributive property from area | 3 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 143 · P4

Say: Read P4. Give time to try before comparing methods.

Ask: P4: Which two quantities does the outside factor multiply?

Expect: P4: 4(x + 3) = 4x + 12 square units

If stuck: P4: Split the rectangle or repeated group into both parts. Count the outside factor on each part.

Slide 324. Combine like terms | 3 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 143 · P5, P6

Say: Read P5 and P6. Give time to try before comparing methods.

Ask: P5: How many groups of the same unknown are being added? P6: Which two quantities does the outside factor multiply?

Expect: P5: 5x; 15 when x = 3; P6: No: 12 ≠ 6. The equivalent form is 4t + 8.

If stuck: P5: Draw one box for each x, then count the boxes. Adding boxes does not change them into squared units. P6: Split the rectangle or repeated group into both parts. Count the outside factor on each part.

Slide 325. Watch the thinking | 3 min

Topic: Expressions and equivalent forms
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 18

If stuck: Return to the matching unit in the model.

Slide 326. Your turn | 3 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 144 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 12; both forms give 22

If stuck: Compare with the previous example without copying its numbers.

Slide 327. Two ways to see it | 8 min

Topic: Expressions and equivalent forms
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Partner compare: How does each smaller rectangle match one term? Agree or disagree: Does one matching input prove two expressions are equivalent?

Expect: Accept drawings and equations that preserve the same quantities and units. Partitions the rectangle and counts both areas.

If stuck: Ask which area matches each term.

Slide 328. A claim to check | 7 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 144 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: 5z + 10. Each of the five groups contains two extra units.

If stuck: Draw five equal rows, each with z and 2.

Slide 329. Words and meaning | 4 min

Topic: Expressions and equivalent forms
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: In 5n, n can be 2 or 7. 5n + 3 4^2 means 4 × 4. 2(x + 5) and 2x + 10

If stuck: Read one definition and draw its example.

Slide 330. In your own words | 3 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 141 · S

Say: Use your own words and one example.

Ask: Partner compare: How does each smaller rectangle match one term?

Expect: An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.

If stuck: Point to the model and name what each part represents.

Slide 331. Exit ticket | 8 min

Topic: Expressions and equivalent forms
Student: Day 036 · Workbook p. 145 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 6x + 12

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 332. The week ahead | 5 min

Topic: Expressions and equivalent forms
Student: Days 038-042 · Workbook pp. 156-165

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 333. Tutor reference

Topic: Expressions and equivalent forms
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 6x + 12

Got it: Selects 6x + 12 and explains that 3 multiplies both 2x and 4.

Almost there: Selects 6x + 12 but gives an incomplete explanation or one arithmetic slip in the check.

Needs re-teaching: Selects 6x + 4 or another form and cannot show both products.

Sunday. Equations and related quantities

Day 037, pages 146-155 | PowerPoint slides 334-352

Goal: Solve equations by keeping both sides equal and connect a rule to a table and graph.

Set out: paper strips, counters, grid paper.

Check first: Equivalent expressions and inverse operations.

Pacing: Retrieval 5; launch 5; explore 15; discuss 12; synthesis 5; mixed review 10; exit 8; preview 5. Total 65 min.

Explore: what to watch for

Removes or divides equally on both sides. Why it matters: Preserves equality. Tutor move: Ask the student to state why the balance remains level.

Splits the fractional whole into equal parts. Why it matters: Keeps the unknown whole visible. Tutor move: Ask which parts the given amount represents.

Uses a tape or inverse operations and checks. Why it matters: Both methods preserve the same relationship. Tutor move: Accept either and connect each step to the other.

Likely misconceptions

Subtracts the coefficient. Why: Does not read juxtaposition as multiplication. Tutor move: Read 4b as four equal groups of b.

Changes one side only. Why: Loses equality. Tutor move: Use matching actions on a balance drawing.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 334. Week 6, Sunday

Topic: Equations and related quantities
Student: Watch and discuss

Say: Solve equations by keeping both sides equal and connect a rule to a table and graph.

Ask: What do you already know about this idea?

Expect: Equivalent expressions and inverse operations.

If stuck: Ask the student to state why the balance remains level.

Slide 335. Today

Topic: Equations and related quantities
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: A solution makes the equation true. The same operation on both sides preserves equality.

If stuck: Read the learning goal together.

Slide 336. Recall | 3 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 146 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 42 points; 2: 1 meters; 3: 40; 4: 0, 12, and 21 centimeters

If stuck: Give time to draw before giving a hint.

Slide 337. Recall: answers | 2 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 146 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 42 points; 1 meters; 40; 0, 12, and 21 centimeters

If stuck: Rebuild one missed item with its model.

Slide 338. Additive equation with whole numbers | 5 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 147 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: P1: What operation will undo the addition while keeping both sides equal?

Expect: P1: $x = 12$

If stuck: P1: Draw the total as an unknown part plus a known part. Remove the same known amount from each side.

Slide 339. Multiplicative equation with whole numbers | 3 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 147 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: P2: How many equal groups or equal fractional parts make the known total?

Expect: P2: $4b = 28$; $b = 7$ marbles; P3: $x = 4.5$

If stuck: P2: Use equal boxes or a tape. Find one part first, then build the whole unknown if needed.

Slide 340. The model | 3 min

Topic: Equations and related quantities
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Four equal unknown groups balance 28.

If stuck: Ask the student to state why the balance remains level.

Slide 341. Fractional coefficient | 3 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 148 · P4, P5

Say: Read P4 and P5. Give time to try before comparing methods.

Ask: P4: How many equal groups or equal fractional parts make the known total? P5: Which quantities must add or multiply to the total in the story?

Expect: P4: x = 15; P5: a + 6.50 = 18; a = $11.50

If stuck: P4: Use equal boxes or a tape. Find one part first, then build the whole unknown if needed. P5: Label the unknown with its unit. Draw a tape that matches the story before solving.

Slide 342. Equation | 2 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 148 · P6

Say: Read P6. Give time to try before comparing methods.

Ask: P6: Which quantity is the input? What rule gives the output?

Expect: P6: (0, 0), (1, 2.5), (2, 5), (4, 10). Input n; output c.

If stuck: P6: Label the horizontal axis with the input. Build one table row at a time before plotting.

Slide 343. Watch the thinking | 2 min

Topic: Equations and related quantities
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: x = 5.25

If stuck: Return to the matching unit in the model.

Slide 344. Your turn | 2 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 149 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 5; 7

If stuck: Compare with the previous example without copying its numbers.

Slide 345. Two ways to see it | 6 min

Topic: Equations and related quantities
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Partner compare: Which part of your tape shows 10 in $(\frac{2}{3})x = 10$? Agree or disagree: Does a solution have to work when we put it back into the original equation?

Expect: Accept drawings and equations that preserve the same quantities and units. Splits the fractional whole into equal parts.

If stuck: Ask which parts the given amount represents.

Slide 346. A claim to check | 6 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 149 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: x = 7. Six equal groups total 42.

If stuck: Draw six equal boxes with a total of 42.

Slide 347. Words and meaning | 3 min

Topic: Equations and related quantities
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: x + 7 = 19 12 solves x + 7 = 19. Number of notebooks, n. Total cost, c = 2.5n.

If stuck: Read one definition and draw its example.

Slide 348. Mixed review | 10 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 150 · M1, M2, M3

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: 80 shells; 3; 8.528

If stuck: Name the quantities and units before calculating.

Slide 349. In your own words | 2 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 146 · S

Say: Use your own words and one example.

Ask: Partner compare: Which part of your tape shows 10 in $(\frac{2}{3})x = 10$?

Expect: A solution makes the equation true. The same operation on both sides preserves equality.

If stuck: Point to the model and name what each part represents.

Slide 350. Exit ticket | 8 min

Topic: Equations and related quantities
Student: Day 037 · Workbook p. 151 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: $x = 20$

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 351. The week ahead | 5 min

Topic: Equations and related quantities
Student: Days 038-042 · Workbook pp. 156-165

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 352. Tutor reference

Topic: Equations and related quantities
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: $x = 20$

Got it: Finds 20 and checks $(\frac{3}{5}) \times 20 = 12$, with equal parts or a valid inverse operation.

Almost there: Finds 20 using a valid method but omits the check or makes one calculation slip after a correct setup.

Needs re-teaching: Multiplies 12 by $\frac{3}{5}$, subtracts $\frac{3}{5}$, or cannot connect 12 to three parts.

PowerPoint answer review

After Friday, outside the weekend lesson time. Say: Check one completed item at a time. Ask: Where does your method first differ from this one? Expect: The answers and worked reasoning printed in the complete keys below. If stuck: Rebuild the quantity with its model, then have the student repair the first incorrect step.

Keep these slides out of view during independent work. The index below follows the answer-review slides in order; every answer is also printed in the complete keys.

Slide 353. Week 6 answers

Next lesson: slide 377. Use this review section after Friday.

Slide 354. Expressions and equivalent forms (1/4)

Items: S11:R1; S11:R2; S11:R3; S11:R4. Follow along: Day 036 · Workbook p. 141 · Items R1, R2, R3, R4

Slide 355. Expressions and equivalent forms (2/4)

Items: S11:S; S11:P1; S11:P2; S11:P3. Follow along: Day 036 · Workbook pp. 141, 142 · Items S, P1, P2, P3

Slide 356. Expressions and equivalent forms (3/4)

Items: S11:P4; S11:P5; S11:P6; S11:F; S11:D. Follow along: Day 036 · Workbook pp. 143, 144 · Items P4, P5, P6, F, D

Slide 357. Expressions and equivalent forms (4/4)

Items: S11:E. Follow along: Day 036 · Workbook p. 145 · Items E

Slide 358. Equations and related quantities (1/4)

Items: S12:R1; S12:R2; S12:R3; S12:R4. Follow along: Day 037 · Workbook p. 146 · Items R1, R2, R3, R4

Slide 359. Equations and related quantities (2/4)

Items: S12:S; S12:P1; S12:P2; S12:P3. Follow along: Day 037 · Workbook pp. 146, 147 · Items S, P1, P2, P3

Slide 360. Equations and related quantities (3/4)

Items: S12:P4; S12:P5; S12:P6; S12:F; S12:D. Follow along: Day 037 · Workbook pp. 148, 149 · Items P4, P5, P6, F, D

Slide 361. Equations and related quantities (4/4)

Items: S12:M1; S12:M2; S12:M3; S12:E. Follow along: Day 037 · Workbook pp. 150, 151 · Items M1, M2, M3, E

Slide 362. Study guide: Try it

Items: W06_Study_Try:1; W06_Study_Try:2; W06_Study_Try:3; W06_Study_Try:4; W06_Study_Try:5. Follow along: Day 037 · Workbook p. 155 · Items 1, 2, 3, 4, 5

Slide 363. Daily practice (1/3)

Items: Day038_Practice:1; Day038_Practice:2; Day038_Practice:3; Day038_Practice:4; Day038_Practice:5. Follow along: Day 038 · Workbook p. 156 · Items 1, 2, 3, 4, 5

Slide 364. Daily practice (2/3)

Items: Day038_Practice:6; Day038_Practice:7; Day038_Practice:8; Day038_Practice:9; Day038_Practice:10. Follow along: Day 038 · Workbook pp. 156, 157 · Items 6, 7, 8, 9, 10

Slide 365. Daily practice (3/3)

Items: Day038_Practice:11; Day038_Practice:12; Day038_Practice:13. Follow along: Day 038 · Workbook p. 157 · Items 11, 12, 13

Slide 366. Daily practice (1/3)

Items: Day039_Practice:1; Day039_Practice:2; Day039_Practice:3; Day039_Practice:4; Day039_Practice:5. Follow along: Day 039 · Workbook p. 158 · Items 1, 2, 3, 4, 5

Slide 367. Daily practice (2/3)

Items: Day039_Practice:6; Day039_Practice:7; Day039_Practice:8; Day039_Practice:9; Day039_Practice:10. Follow along: Day 039 · Workbook pp. 158, 159 · Items 6, 7, 8, 9, 10

Slide 368. Daily practice (3/3)

Items: Day039_Practice:11; Day039_Practice:12; Day039_Practice:13. Follow along: Day 039 · Workbook p. 159 · Items 11, 12, 13

Slide 369. Daily practice (1/3)

Items: Day040_Practice:1; Day040_Practice:2; Day040_Practice:3; Day040_Practice:4; Day040_Practice:5. Follow along: Day 040 · Workbook p. 160 · Items 1, 2, 3, 4, 5

Slide 370. Daily practice (2/3)

Items: Day040_Practice:6; Day040_Practice:7; Day040_Practice:8; Day040_Practice:9; Day040_Practice:10. Follow along: Day 040 · Workbook pp. 160, 161 · Items 6, 7, 8, 9, 10

Slide 371. Daily practice (3/3)

Items: Day040_Practice:11; Day040_Practice:12; Day040_Practice:13. Follow along: Day 040 · Workbook p. 161 · Items 11, 12, 13

Slide 372. Daily practice (1/3)

Items: Day041_Practice:1; Day041_Practice:2; Day041_Practice:3; Day041_Practice:4; Day041_Practice:5. Follow along: Day 041 · Workbook p. 162 · Items 1, 2, 3, 4, 5

Slide 373. Daily practice (2/3)

Items: Day041_Practice:6; Day041_Practice:7; Day041_Practice:8; Day041_Practice:9; Day041_Practice:10. Follow along: Day 041 · Workbook pp. 162, 163 · Items 6, 7, 8, 9, 10

Slide 374. Daily practice (3/3)

Items: Day041_Practice:11; Day041_Practice:12; Day041_Practice:13; Day041_Practice:14. Follow along: Day 041 · Workbook p. 163 · Items 11, 12, 13, 14

Slide 375. Weekly check (1/2)

Items: Day042_Practice:1; Day042_Practice:2; Day042_Practice:3; Day042_Practice:4; Day042_Practice:5. Follow along: Day 042 · Workbook p. 164 · Items 1, 2, 3, 4, 5

Slide 376. Weekly check (2/2)

Items: Day042_Practice:6; Day042_Practice:7; Day042_Practice:8; Day042_Practice:9; Day042_Practice:10. Follow along: Day 042 · Workbook p. 165 · Items 6, 7, 8, 9, 10

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 038. Monday | pages 156-157

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Expressions and equations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 039. Tuesday | pages 158-159

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Expressions and equations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 040. Wednesday | pages 160-161

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 041. Thursday | pages 162-163

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Decimal practice: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 042. Friday | pages 164-165

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Expressions and equivalent forms

Day 036, student workbook pages 141-145

R1 [R-RATE] | student p. 141

Answer: 0.65 dollars per marker

Reasoning: Use the named units, calculate, then check with the inverse operation.

R2 [F-DIV] | student p. 141

Answer: 6

Reasoning: Rename each whole in thirds, then count groups of two thirds.

Watch for $2\frac{2}{3}$: Multiplying by the divisor answers a different question.

R3 [N-DIV] | student p. 141

Answer: 12

Reasoning: Use long division and verify quotient × 24 equals the dividend.

R4 [P-WHOLE] | student p. 141

Answer: 60 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 5.40 shells: Taking 30% of the known part does not recover the whole.

S [EXPLAIN] | student p. 141

Answer: Answers vary. An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [E-EXPR] | student p. 142

Answer: 5n; $10; $20

Reasoning: Five equal costs give 5n. Substitute each sheet price.

Watch for n = 5: Confuses the number of sheets with the price.

P2 [E-EVAL] | student p. 142

Answer: 17

Reasoning: 2 × 6 + 5 = 12 + 5 = 17.

Watch for 22: Adds before multiplying.

P3 [E-EXP] | student p. 142

Answer: 16; 8; 27; $\frac{4}{9}$. 4^2 and 4×2 have different meanings.

Reasoning: Write repeated factors: 4×4, 4×2, $3 \times 3 \times 3$, and $\frac{2}{3} \times \frac{2}{3}$.

Watch for $4^2 = 8$: Treats the exponent as a factor.

P4 [E-DIST] | student p. 143

Answer: $4(x + 3) = 4x + 12$ square units

Reasoning: The two smaller areas are $4x$ and 4×3. Their sum is $4x + 12$.

Watch for $4x + 3$: Does not multiply the second width by the height.

P5 [E-LIKE] | student p. 143

Answer: $5x$; 15 when $x = 3$

Reasoning: There are five groups of x.

Watch for x□: Confuses adding equal terms with multiplying equal factors.

P6 [E-DIST] | student p. 143

Answer: No: $12 \neq 6$. The equivalent form is $4t + 8$.

Reasoning: One counterexample proves the forms differ. Distribution gives $4t + 8$.

Watch for Yes, both have 4t and 2: Matches symbols without comparing quantities.

F [E-DIST] | student p. 144

Answer: 12; both forms give 22

Reasoning: Use the worked example in the slides to check each step.

D [E-DIST] | student p. 144

Answer: $5z + 10$. Each of the five groups contains two extra units.

Reasoning: Lena multiplied one part only. Draw five equal rows, each with z and 2.

E [E-DIST] | student p. 145

Answer: $6x + 12$

Reasoning: $3 \times 2x = 6x$ and $3 \times 4 = 12$.

Watch for $6x + 4$: Distributes only to the variable term.

Equations and related quantities

Day 037, student workbook pages 146-151

R1 [P-PART] | student p. 146

Answer: 42 points

Reasoning: Take all 30 points, then add forty percent, which is 12 points.

R2 [F-DIV] | student p. 146

Answer: 1 meters

Reasoning: Divide area by width. Check that length times width equals area.

Watch for $\frac{9}{16}$: Multiplying by the divisor answers a different question.

R3 [D-DIV] | student p. 146

Answer: 40

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.4: Changing only the divisor changes the quotient. Scale both quantities by 100.

R4 [R-TABLE] | student p. 146

Answer: 0, 12, and 21 centimeters

Reasoning: Multiply each time by the unit rate. Zero time matches zero distance.

S [EXPLAIN] | student p. 146

Answer: Answers vary. A solution makes the equation true. The same operation on both sides preserves equality.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [E-ADD] | student p. 147

Answer: x = 12

Reasoning: Subtract 7 from each side.

Watch for x = 26: Adds instead of undoing the addition.

P2 [E-MULT] | student p. 147

Answer: 4b = 28; b = 7 marbles

Reasoning: Split 28 into four equal groups.

Watch for b = 24: Subtracts the coefficient.

P3 [E-MULT] | student p. 147

Answer: x = 4.5

Reasoning: Divide both sides by 1.2. Then 1.2 × 4.5 = 5.4.

Watch for x = 4.2: Subtracts 1.2 from 5.4.

P4 [E-MULT] | student p. 148

Answer: x = 15

Reasoning: Two parts total 10, so one part is 5. All three parts total 15.

Watch for x = $\frac{20}{3}$: Multiplies by the coefficient instead of finding the whole.

P5 [E-STORY] | student p. 148

Answer: a + 6.50 = 18; a = $11.50

Reasoning: The saved amount plus the extra amount equals the price.

Watch for a = $24.50: Combines the given amounts without matching their roles.

P6 [E-RELATE] | student p. 148

Answer: (0, 0), (1, 2.5), (2, 5), (4, 10). Input n; output c.

Reasoning: Multiply each notebook count by 2.5. The chosen count determines the cost.

Watch for (2.5, 1) for one notebook: Reverses the input and output.

F [E-MULT] | student p. 149

Answer: 5; 7

Reasoning: Use the worked example in the slides to check each step.

D [E-MULT] | student p. 149

Answer: x = 7. Six equal groups total 42.

Reasoning: Treats multiplication as addition. Draw six equal boxes with a total of 42.

M1 [P-WHOLE] | student p. 150

Answer: 80 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 7.20 shells: Taking 30% of the known part does not recover the whole.

M2 [F-DIV] | student p. 150

Answer: 3

Reasoning: Rename each whole in thirds, then count groups of two thirds.

Watch for 1 $\frac{1}{3}$: Multiplying by the divisor answers a different question.

M3 [D-ADD] | student p. 150

Answer: 8.528

Reasoning: Line up decimal points. Add digits in the same place-value columns.

E [E-MULT] | student p. 151

Answer: x = 20

Reasoning: Three fifths total 12, so one fifth is 4. Five fifths total 20.

Watch for x = $\frac{36}{5}$: Multiplies 12 by $\frac{3}{5}$ instead of finding the whole.

Study guide: Try it

Day 037, student workbook pages 152-155

1 [E-DIST] | student p. 155

Answer: 2x + 14

Reasoning: Multiply x and 7 by 2.

Watch for 2x + 7: Multiplies only the first term.

2 [E-DIST] | student p. 155

Answer: 3x + 18

Reasoning: Multiply x and 6 by 3.

Watch for 3x + 6: Multiplies only the first term.

3 [E-ADD] | student p. 155

Answer: x = 14

Reasoning: Subtract 9 from both sides.

4 [E-ADD] | student p. 155

Answer: x = 17

Reasoning: Subtract 14 from both sides.

5 [E-ADD] | student p. 155

Answer: x = 7

Reasoning: Subtract 11 from both sides.

Daily practice

Day 038, student workbook pages 156-157

1 [N-FACT] | student p. 156

Answer: 24

2 [N-FACT] | student p. 156

Answer: 56

3 [N-FACT] | student p. 156

Answer: 32

4 [N-FACT] | student p. 156

Answer: 72

5 [F-DIV] | student p. 156

Answer: $1\ \frac{1}{3}$ meters

Reasoning: Divide area by width. Check that length times width equals area.

Watch for $\frac{3}{4}$: Multiplying by the divisor answers a different question.

6 [F-ADD] | student p. 156

Answer: $\frac{11}{12}$

Reasoning: Rename both fractions with a common denominator. Add the counts of equal parts.

7 [N-FACT] | student p. 156

Answer: 8

8 [E-EVAL] | student p. 156

Answer: 17

Reasoning: $2 \times 5 + 7 = 17$.

9 [E-DIST] | student p. 157 | Built-in review: Expressions and equations

Answer: 2x + 14

Reasoning: Multiply x and 7 by 2.

Watch for 2x + 7: Multiplies only the first term.

10 [E-EXP] | student p. 157 | Built-in review: Expressions and equations

Answer: 25

Reasoning: Multiply 2 factors of 5.

Watch for 10: Multiplies the base by the exponent.

11 [E-EVAL] | student p. 157 | Built-in review: Expressions and equations

Answer: 11

Reasoning: $4 \times 2 + 3 = 11$.

12 [E-EXPR] | student p. 157 | Built-in review: Expressions and equations

Answer: 3b; 12 beads

Reasoning: 3 equal groups of b give 3b.

13 [E-DIST] | student p. 157

Answer: 5z + 10. Each of the five groups contains two extra units.

Watch for 5z + 2: Lena multiplied one part only.

Daily practice

Day 039, student workbook pages 158-159

1 [F-EQUIV] | student p. 158

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 158

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 158

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 158

Answer: 0.125; 12.5%

5 [F-DIV] | student p. 158

Answer: $1\ \frac{3}{7}$

Reasoning: Keep four sevenths and multiply by five halves. The check recovers four sevenths.

6 [P-WHOLE] | student p. 158

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

7 [N-FACT] | student p. 158

Answer: 12

8 [E-MULT] | student p. 158

Answer: x = 25

Reasoning: 2 equal parts total 10. Find one part, then all 5.

9 [E-ADD] | student p. 159 | Built-in review: Expressions and equations

Answer: x = 7

Reasoning: Subtract 11 from both sides.

10 [E-MULT] | student p. 159 | Built-in review: Expressions and equations

Answer: x = 8

Reasoning: Divide both sides by 7.

Watch for 49: Subtracts instead of dividing.

11 [E-MULT] | student p. 159 | Built-in review: Expressions and equations

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

12 [E-MULT] | student p. 159 | Built-in review: Expressions and equations

Answer: x = 16

Reasoning: 3 equal parts total 12. Find one part, then all 4.

13 [E-MULT] | student p. 159

Answer: x = 7. Six equal groups total 42.

Watch for x = 36: Treats multiplication as addition.

Daily practice

Day 040, student workbook pages 160-161

1 [N-SCALE] | student p. 160

Answer: 82

2 [N-SCALE] | student p. 160

Answer: 8

3 [N-SCALE] | student p. 160

Answer: 17

4 [N-SCALE] | student p. 160

Answer: 24

5 [E-MULT] | student p. 160

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

6 [E-DIST] | student p. 160

Answer: 4x + 20

Reasoning: Multiply x and 5 by 4.

Watch for 4x + 5: Multiplies only the first term.

7 [D-MULT] | student p. 160

Answer: 0.0050

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.0500: The decimal point is one place too far right. Check the product against both factors.

8 [N-DIV] | student p. 160

Answer: 12

Reasoning: Use long division and verify quotient × 24 equals the dividend.

9 [P-WHOLE] | student p. 161

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

10 [R-EQUIV] | student p. 161

Answer: 15 blue tiles

Reasoning: The number of batches is 9 ÷ 3. Multiply that count by 5.

Watch for 11 blue tiles: Adding the same amount to both counts changes the ratio.

11 [N-FACT] | student p. 161

Answer: 24

12 [G-VOL] | student p. 161

Answer: 72 cm^3

13 [E-STORY] | student p. 161 | Built-in review: Weekly puzzle

Answer: 6 and 11

Reasoning: Remove the extra 5: 17 − 5 = 12. Split the rest equally: 6 and 6. Restore 5 to one number.

Watch for 5 and 12: Uses the difference as one of the numbers.

Daily practice

Day 041, student workbook pages 162-163

1 [N-EST] | student p. 162

Answer: 800

2 [N-EST] | student p. 162

Answer: 9

3 [N-EST] | student p. 162

Answer: About 100

4 [N-EST] | student p. 162

Answer: About 20

5 [E-MULT] | student p. 162

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

6 [E-DIST] | student p. 162

Answer: 4x + 20

Reasoning: Multiply x and 5 by 4.

Watch for 4x + 5: Multiplies only the first term.

7 [E-MULT] | student p. 162

Answer: x = 8

Reasoning: Divide both sides by 7.

Watch for 49: Subtracts instead of dividing.

8 [F-DIV] | student p. 162

Answer: $2\frac{1}{2}$; stories vary

Reasoning: Example: how many one-third-meter lengths fit in five sixths meter, allowing part of a length?

9 [D-DIV] | student p. 163 | Built-in review: Decimal practice

Answer: 1.2

Reasoning: Ninety-six tenths shared among eight gives twelve tenths.

10 [D-DIV] | student p. 163 | Built-in review: Decimal practice

Answer: 40

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.4: Changing only the divisor changes the quotient. Scale both quantities by 100.

11 [D-MONEY] | student p. 163 | Built-in review: Decimal practice

Answer: $2.57

Reasoning: The cost is $7.43. Subtract that from $10.00.

12 [D-SUB] | student p. 163 | Built-in review: Decimal practice

Answer: 8.714

Reasoning: Write 10.000 and regroup through the zero columns.

13 [E-EXPR] | student p. 163

Answer: 6b; 24 beads

Reasoning: 6 equal groups of b give 6b.

14 [E-MULT] | student p. 163

Answer: x = 7. Six equal groups total 42.

Watch for x = 36: Treats multiplication as addition.

Weekly check

Day 042, student workbook pages 164-165

1 [E-MULT] | student p. 164

Answer: x = 3.2

Reasoning: Divide both sides by 1.5; then check the product.

2 [E-DIST] | student p. 164

Answer: 4x + 20

Reasoning: Multiply x and 5 by 4.

Watch for 4x + 5: Multiplies only the first term.

3 [E-MULT] | student p. 164

Answer: x = 8

Reasoning: Divide both sides by 7.

Watch for 49: Subtracts instead of dividing.

4 [E-EVAL] | student p. 164

Answer: 13

Reasoning: 3 × 4 + 1 = 13.

5 [D-SUB] | student p. 164

Answer: 8.714

Reasoning: Write 10.000 and regroup through the zero columns.

6 [D-MULT] | student p. 165

Answer: 0.0100

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.1000: The decimal point is one place too far right. Check the product against both factors.

7 [D-MULT] | student p. 165

Answer: 0.0050

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.0500: The decimal point is one place too far right. Check the product against both factors.

8 [P-WHOLE] | student p. 165

Answer: 40 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 3.60 shells: Taking 30% of the known part does not recover the whole.

9 [R-EQUIV] | student p. 165

Answer: 15 blue tiles

Reasoning: The number of batches is 9 ÷ 3. Multiply that count by 5.

Watch for 11 blue tiles: Adding the same amount to both counts changes the ratio.

10 [E-DIST] | student p. 165

Answer: 5z + 10. Each of the five groups contains two extra units.

Watch for 5z + 2: Lena multiplied one part only.